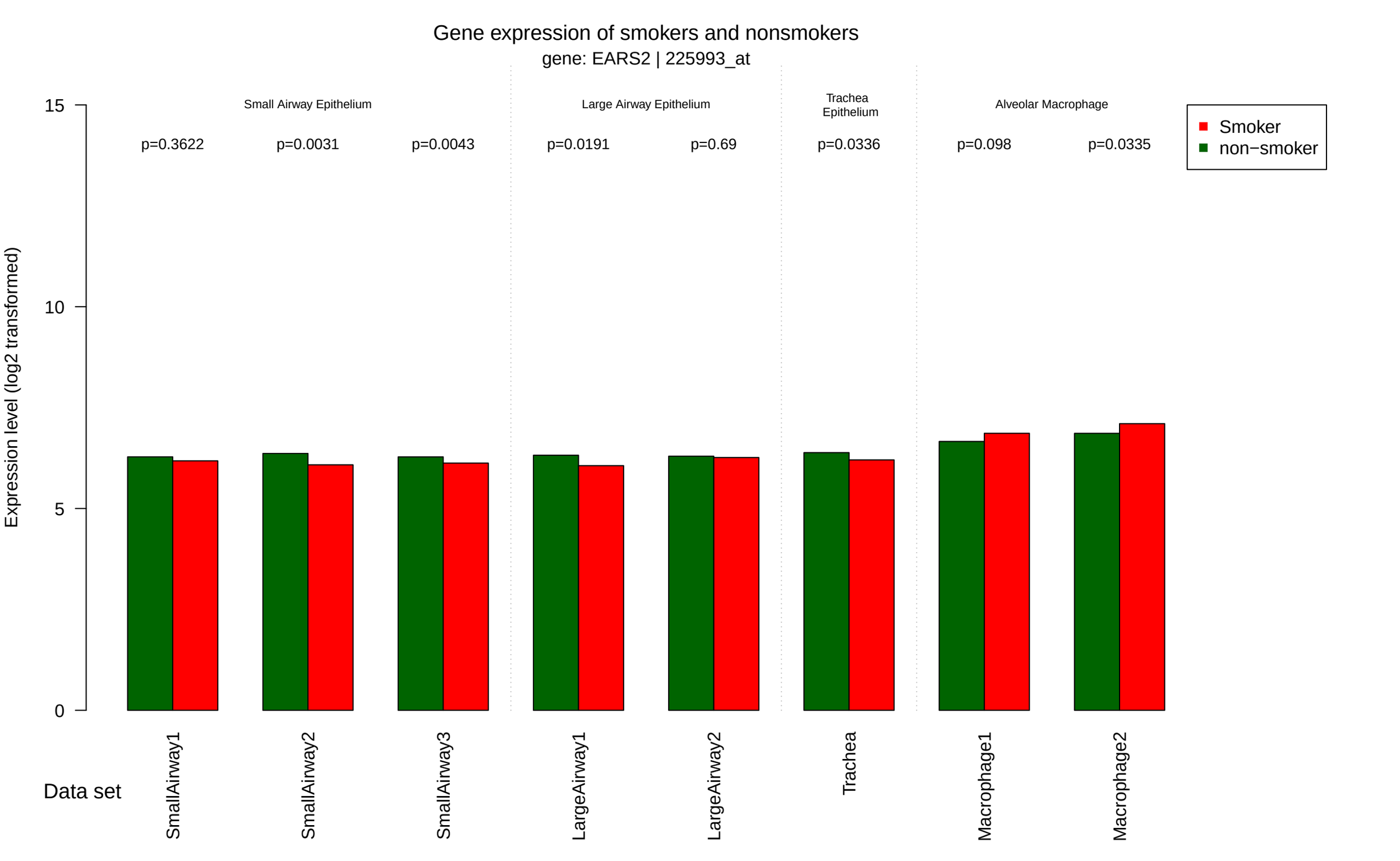
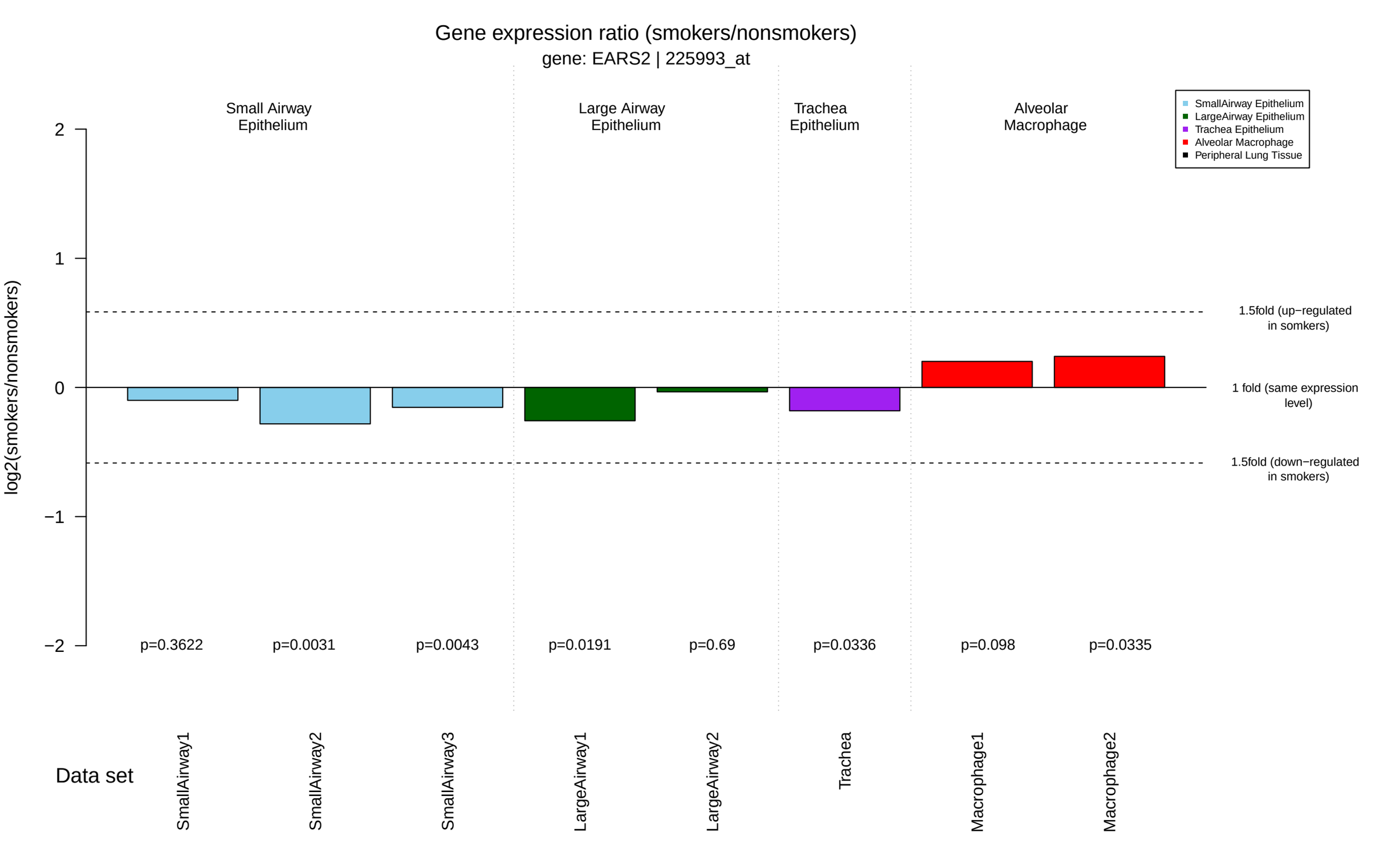
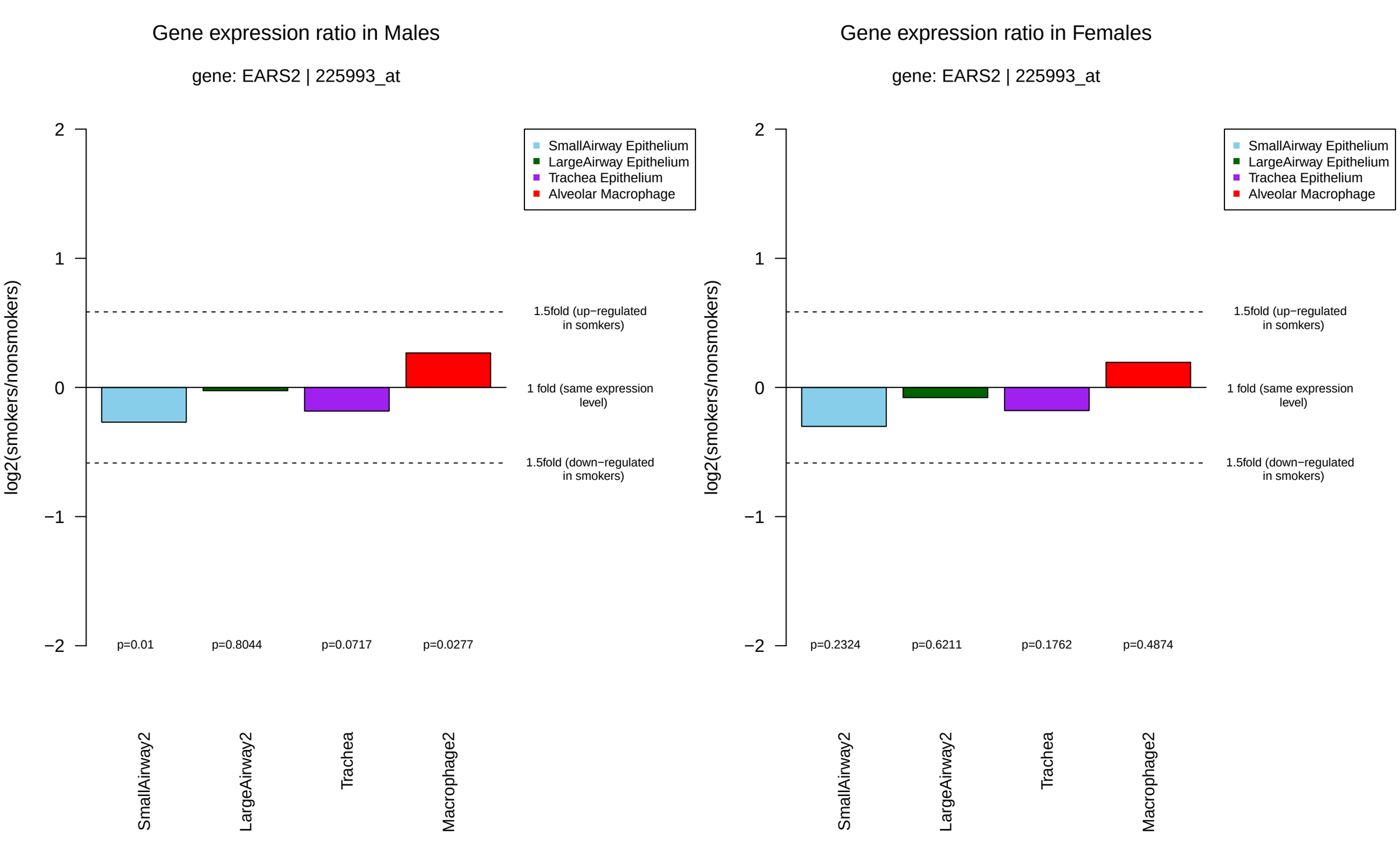
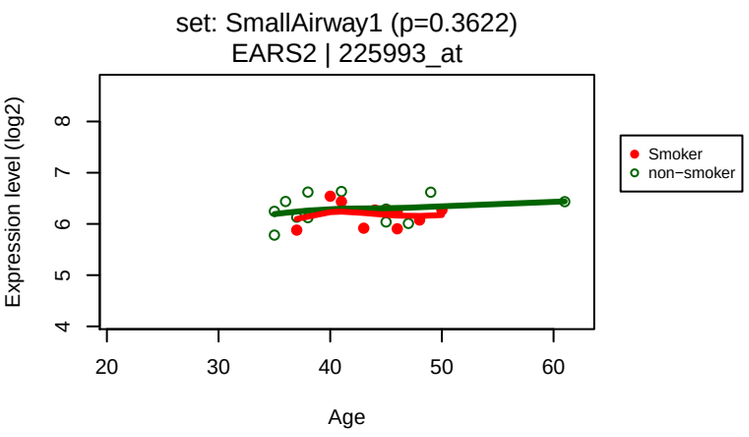
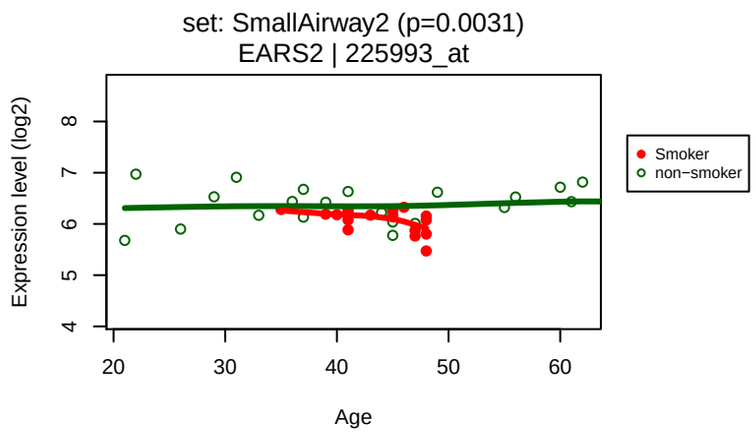
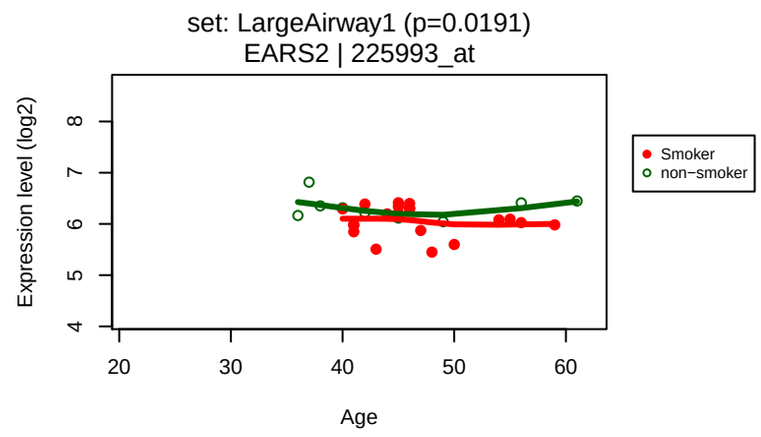

Gene expression of smokers and nonsmokers
gene: EARS2 | 225993_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3622
p=0.0031
p=0.0043
p=0.0191
p=0.69
p=0.0336
p=0.098
p=0.0335
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

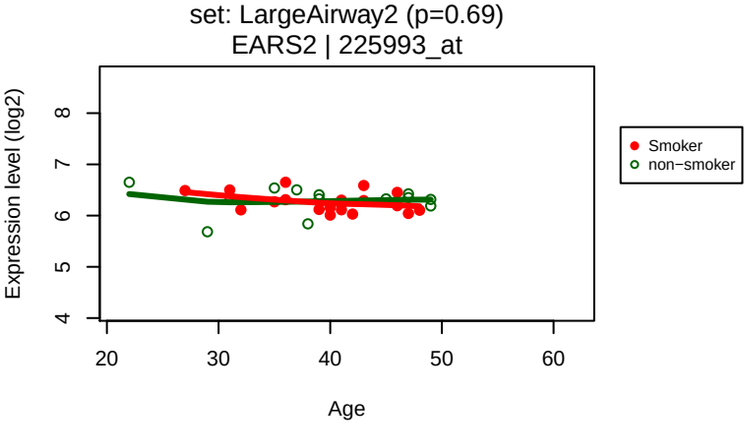

Gene expression ratio (smokers/nonsmokers)
gene: EARS2 | 225993_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3622
p=0.0031
p=0.0043
p=0.0191
p=0.69
p=0.0336
p=0.098
p=0.0335
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

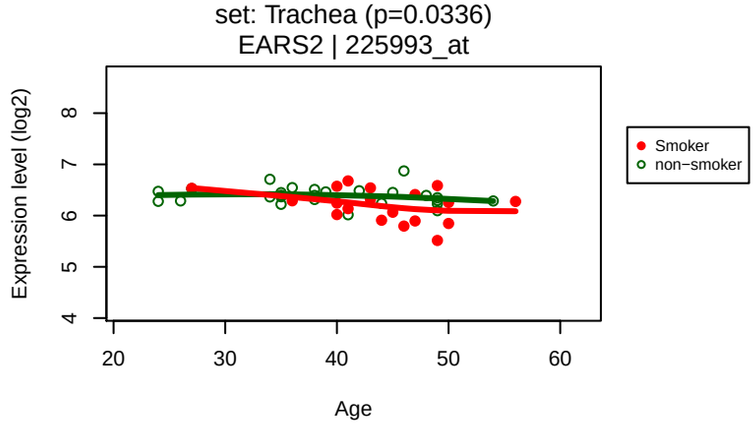

Gene expression ratio in Males
gene: EARS2 | 225993_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.01
p=0.8044
p=0.0717
p=0.0277
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EARS2 | 225993_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2324
p=0.6211
p=0.1762
p=0.4874
SmallAirway2
LargeAirway2
Trachea
Macrophage2

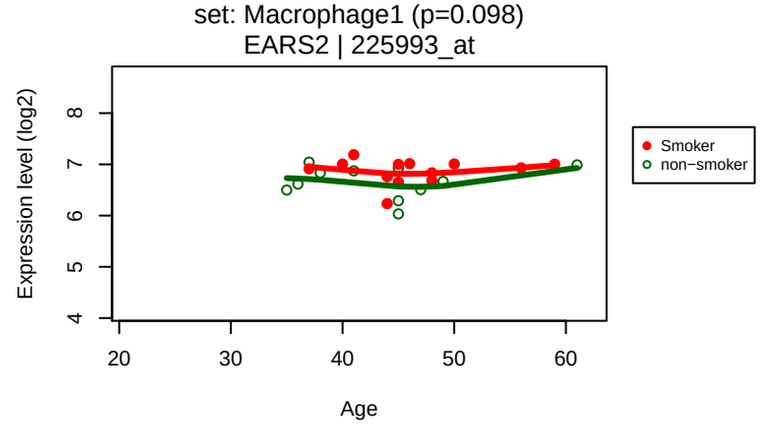

set: SmallAirway1 (p=0.3622)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker

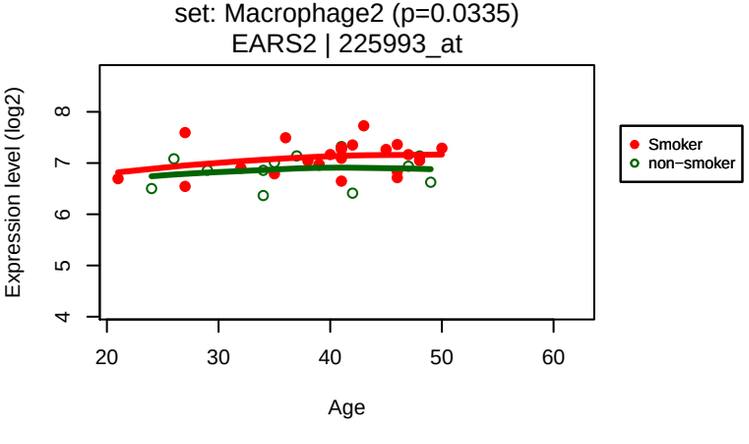

set: SmallAirway2 (p=0.0031)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker

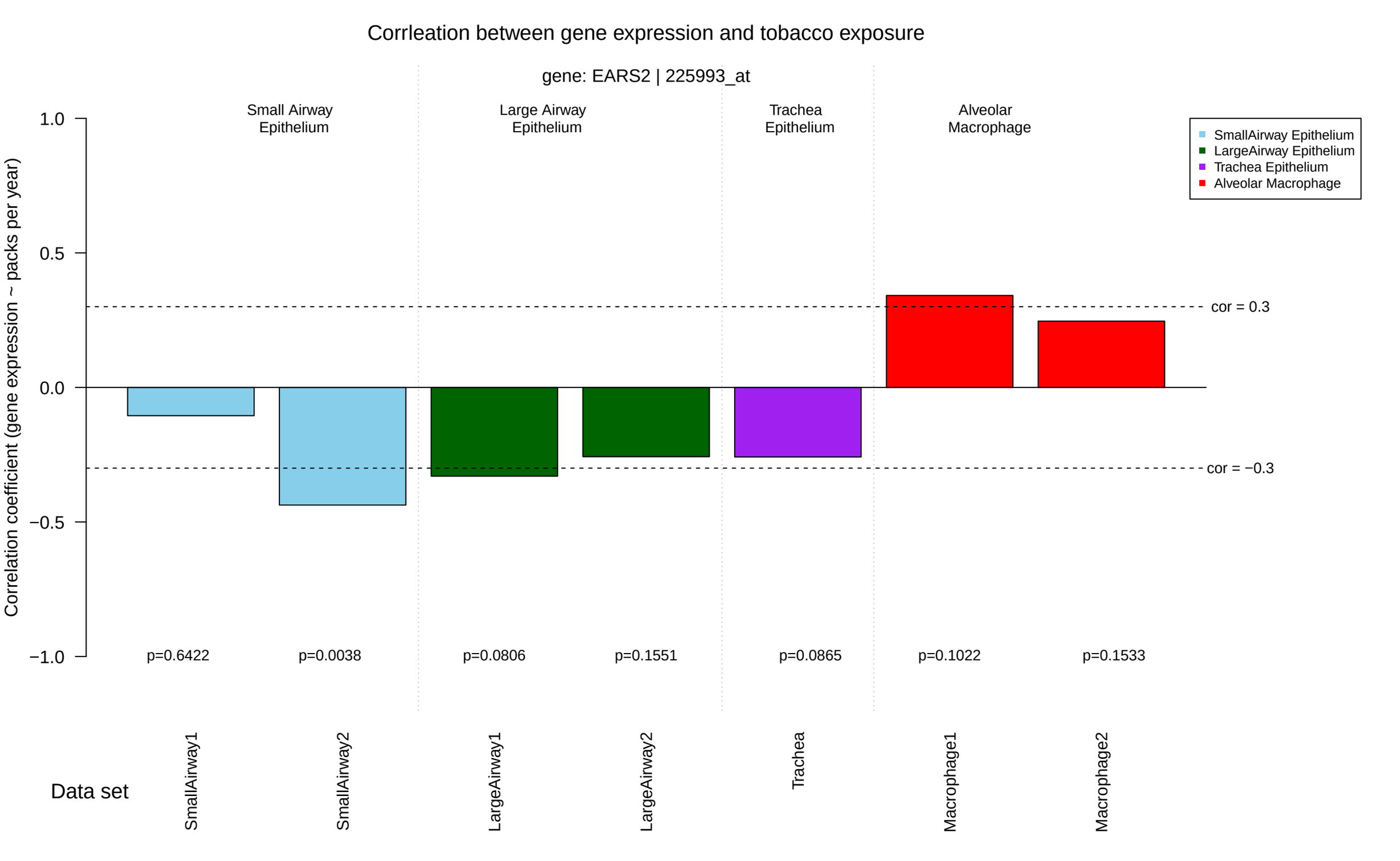

set: LargeAirway1 (p=0.0191)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.69)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0336)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.098)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0335)
EARS2 | 225993_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EARS2 | 225993_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6422
p=0.0038
p=0.0806
p=0.1551
p=0.0865
p=0.1022
p=0.1533
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2